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To the Editors,
Array

Dear Editors,

I am submitting the manuscript “WAGER: Within-cell Antisymmetric Gain Evaluation of Resolution” for consideration as a regular paper in *Array*.

The paper addresses a measurement problem that recurs across machine learning: when a benchmark’s labels are strongly predictable from a known nuisance feature, a new model can raise its headline score by fitting that regularity more precisely rather than by predicting individual examples better. Scene graph generation, visual question answering and long-tailed recognition all have this structure. The usual response is to compare a model against a prior-only baseline, but that does not answer the question a benchmark result actually poses: when system B outscores system A, how much of the gain came from the instances and how much from the prior?

WAGER answers that question directly, and the contribution is primarily methodological. Given two frozen probabilistic models, it forms their proper-score contrast, matches test examples sharing a declared prior feature, and crosses their labels. This yields an exact finite-sample identity — the observed gain equals a prior-transported gain plus an antisymmetric instance-alignment gain — whose residual is an order-two U-statistic that, under the quadratic score, equals the within-cell prediction–label covariance the new model has gained, linking it to the classical resolution term of proper-score decomposition. The estimator runs in $O(NK)$ time on cached predictions, fits no baseline model, has no tunable hyperparameter, and comes with cluster-robust intervals and a cell-stratified randomization test. Two further exact results are proved: leave-one-out transport removes an attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in incurs, and coarsening the audit cell shifts the estimate by an explicit between-cell covariance term.

Because the method is a measuring instrument, the paper takes its validation seriously. Beyond recovery and null designs, we verify the coverage of the primary interval in a regime that stresses it, and we report discriminant-validity experiments establishing what the two channels do and do not separate — including an honest limitation, that a monotone recalibration moves score mass between the channels, together with the calibration-matched protocol that removes it and a demonstration that the protocol works.

The empirical results show the instrument earning its keep. On 229,605 Visual Genome relations, a class-only network’s entire improvement over the frequency baseline is provably prior-transported, with exactly zero instance alignment. More strikingly, when a frozen CLIP encoder replaces annotation-derived box geometry, the resulting model scores 0.04655 *lower* overall — an evaluator reading that number would discard it — yet aligns with individual instances significantly *better* than the model that beats it ($+0.00641$, $p \leq .002$), its entire deficit lying in the annotation prior; the advantage survives both post-hoc prior correction and calibration matching. Applied unchanged to long-tailed image classification, the same estimator separates two re-weighting schedules whose aggregate scores hide their composition, and a calibration-matched reanalysis overturns what the raw decomposition appears to say about each of them.

I believe the work suits *Array*’s broad computer-science scope: it is a methodological and

measurement contribution, with proofs, a reference implementation, and validation on two structurally different benchmark families, rather than a new architecture competing for state-of-the-art accuracy. All code, cached model outputs and scripts needed to reproduce every number and figure are publicly available, and the repository URL is given on the title page.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with the journal's policy, the manuscript includes a declaration of generative AI use in its preparation. Author details are supplied on the separate title page, with the manuscript itself anonymized for the review process.

Thank you for your consideration.

Yours sincerely,

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